

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Summer-2023

Course: B. Tech. Branch : Electronics and Computer Engineering Semester : IV

Subject Code & Name: BTECPE405A & Microcontroller & Advanced Processor

Max Marks: 60

Date:22/7/2023

Duration: 3Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Distinguish between Harvard and Von Neumann architecture.	Level 2	6
B) State addressing modes of following instructions of 8051. i. MOV R1, #03H ii. MOV A, R1 iii. MOV A, @R0 iv. MOVC A, @A+DPTR v. ADD A, 40H vi. RL A	Level 3	6
C) Explain the functional pin diagram of 8051 Microcontroller.	Level 2	6
Q.2 Solve Any Two of the following.		12
A) Briefly explain any three unconditional jump instructions of 8051 with suitable examples of each.	Level 1	6
B) Explain the different modes of timer operation in 8051.	Level 1	6
C) Assume that 10 numbers are stored in internal ROM from 1000H to 1009H. Write an assembly language program to find the sum of these numbers.	Level 3	6
Q. 3 Solve Any Two of the following.		12
A) Assume that 8 LEDs are connected to Port 1 of 8051. Write a 8051 C program blink these LEDs continuously.	Level 4	6
B) Write an 8051 program to generate a sine wave using DAC 0808.	Level 1	6
C) Explain the different modes of serial communication in 8051?	Level 2	6
Q.4 Solve Any Two of the following.		12
A) Briefly explain different features of ARM7 architecture.	Level 1	6
B) Provide comprehensive explanation of ARM7 Programmers model.	Level 2	6
C) With suitable example explain concept of 5 stage Pipelining in ARM.	Level 2	6

Q. 5 Solve Any Two of the following.

12

A) Draw and explain memory map of LPC2148.

Level 1

6

B) Explain the features of timers of LPC2148.

Level 1

6

C) Explain the features of RENESAS Microcontroller.

Level 1

6

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